



SKILL DEVELOPMENT LABORATORY-SCENARIO-BASED ASSESSMENT

UNIT II: CASCADING STYLESHEETS

Course Title	WEB TECHNOLOGY
Course Code	ITA02
Assessment Type	Scenario-Based Questions
Duration	90 Minutes (minimum practical working time: 60 minutes)
Maximum Marks	100
Date / Session	

Student Name	MALAI ARASAN	Register No.	192521014
Class / Section		Signature	<i>Malai</i>

General Instructions

- This is an individual practical assessment. Each student must design, implement, test and document the application independently.
- Use only open-source software or freely available open-source development tools.
- The application must run locally in a web browser without requiring a paid service or proprietary web-development platform.
- Maintain HTML and CSS in separate files, except where the task explicitly requires demonstrations of inline and internal CSS.
- Do not copy a complete template. The submitted design, layout decisions and source code must reflect the student's own work.
- The minimum practical working time is 60 minutes. Submit all required source files, screenshots and technical explanations.

Question 1: Responsive University Event Registration Portal Using CSS

Scenario: A university conducts technical workshops, hackathons, cultural events and placement-training programmes. Its existing event-registration webpage is difficult to use because the content is displayed as plain text, event cards are not aligned properly, and the page does not adapt to mobile screens. The university requires a professionally designed, responsive and user-friendly event-registration interface that demonstrates core CSS concepts, normal document flow, advanced layout, positioning and other visual properties.

Task

Design and develop a Responsive University Event Registration Portal using HTML5 and CSS3. The final webpage must be visually consistent, responsive, accessible and properly aligned on desktop and mobile screens.

A. Required Webpage Sections

Section	Minimum Requirement
Header	University name, portal title and navigation menu.
Hero section	A prominent registration message, event date or campaign period, and a call-to-action button.
Event catalogue	At least four event cards representing different categories such as technical, cultural, placement or sports events.
Event-card content	Event title, category, date and time, venue, brief description, registration status and registration button.
Registration form	Student name, register number, department, email address, event selection and submit button.
Footer	Contact information, copyright statement and useful links.

B. Mandatory CSS Requirements

1. Methods of Applying CSS

- Demonstrate inline CSS on one small element only.
- Demonstrate internal CSS for one clearly identified section.
- Use an external style sheet for the main website design.
- In the report, justify which CSS method is most suitable for a multi-page website.

2. Selectors and Core Syntax

- Use valid selector, property and value syntax throughout the style sheet.
- Use at least five selector types: element, class, ID, group, descendant, child, attribute, pseudo-class or pseudo-element.
- Include at least one hover or focus state and one pseudo-element or attribute selector.

3. CSS Box Model

- Every event card must visibly demonstrate content, padding, border and margin.
- Choose one event card and calculate its total horizontal width using the default content-box model.
- Explain how the result changes when box-sizing: border-box is applied.

4. Layout and Normal Flow

- Use Flexbox or CSS Grid for the primary event-card layout.
- Display multiple columns on larger screens and a single column on smaller screens.
- Maintain logical document order so the page remains understandable without advanced positioning.

5. Beyond the Normal Flow

- Demonstrate at least two of the following: relative, absolute, fixed or sticky positioning; z-index; float and clear.
- Suggested applications include a sticky navigation bar, fixed help button or positioned event-status badge.
- Ensure positioned elements do not hide form fields or overlap essential content.

6. Other CSS Properties

- Use at least eight relevant properties from: background colour or image, border-radius, box-shadow, text-shadow, opacity, overflow, cursor, transition, transform, gradient, list-style, visibility and max-width.
- Use transitions or transformations purposefully rather than only for decoration.
- Maintain readable colour contrast and consistent typography.

7. Responsive Design

- Use at least one media query to adapt the navigation, event cards and form to smaller screens.
- Test the page at a desktop viewport and at a mobile viewport of approximately 375 px width.
- Ensure no horizontal scrolling is required in the mobile view.

C. Functional and Quality Expectations

- All navigation links and buttons must have clear labels and visible hover or focus feedback.
- The registration form must use appropriate HTML input types and associated labels.
- The webpage must have consistent alignment, spacing, colour use and visual hierarchy.
- The design must remain readable when browser zoom is increased to 125%.
- The CSS file must be logically organised and include comments for major sections.
- No visible overlap, clipped text, broken cards or unreadable content is permitted.

D. Suggested Open-Source Tools

Tool	Suggested Use
VSCode or Code OSS	Write and organise HTML and CSS files.
Mozilla Firefox	Primary browser testing and responsive-design inspection.
Chromium	Cross-browser testing.
Firefox / Chromium Developer Tools	Inspect selectors, computed styles, box model and responsive layouts.
Live Server or live-server	Run the webpage through a local development server.
Git	Maintain version history and document incremental development.

E. Required Deliverables

1. index.html containing the complete semantic webpage structure.
2. style.css containing the primary external CSS rules.
3. A screenshot of the complete desktop view.
4. A screenshot of the mobile responsive view.
5. A screenshot of the CSS box-model panel for one event card.
6. A one-page technical note explaining the CSS methods, selector types, layout method, positioning technique and responsive behaviour used.
7. The total-width calculation for one event card under the default box model.
8. The complete project folder or a Git repository export containing all files and assets.

F. Required Testing Evidence

- Test the webpage in at least two open-source browsers.
- Test at one desktop and one mobile viewport size.
- Check that navigation, cards, buttons and form controls remain properly aligned.
- Inspect computed styles and the box model using browser Developer Tools.
- Record at least two layout issues identified during testing and state how they were corrected.

G. Suggested Time Allocation

Activity	Suggested Time
HTML structure and semantic sections	10 minutes
Basic CSS, typography and box model	15 minutes
Flexbox or Grid layout	15 minutes
Positioning and additional visual properties	10 minutes
Responsive testing and corrections	10 minutes

Activity	Suggested Time
Screenshots and short documentation	10 minutes
Recommended total	70 minutes

H. Student Submission Checklist

Submission Item	Check
index.html file is included and opens correctly.	☐
style.css file is linked correctly and contains the main CSS rules.	☐
Desktop-view screenshot is included.	☐
Mobile-view screenshot is included.	☐
CSS box-model screenshot is included.	☐
Technical note and event-card width calculation are included.	☐
Testing evidence from two browsers is included.	☐
Complete project folder or repository export is submitted.	☐

I. Evaluation Rubric

Assessment Criterion	Marks	Performance Evidence
Scenario interpretation and content completeness	10	All required sections and content are present and relevant to the university-event scenario.
HTML structure and usability	10	Semantic organisation, labels, navigation and form usability.
CSS syntax, selectors and style-sheet methods	15	Correct syntax, required selector variety and proper use of inline, internal and external CSS.
Box model and visual consistency	15	Correct margins, padding, borders, sizing, alignment, typography and card consistency.
Layout and positioning	15	Effective use of Flexbox or Grid and at least two beyond-normal-flow techniques without overlap.
Responsive design and browser compatibility	15	Effective media queries, desktop/mobile adaptation and testing in two browsers.
Other CSS properties and interaction design	10	Purposeful use of visual properties, hover/focus states, transitions and transformations.
Testing evidence and corrections	5	Developer-tools evidence and documented correction of two layout issues.
Code quality and documentation	5	Readable, organised, commented code and clear technical note.
TOTAL	100	

Academic Integrity Declaration

I declare that the submitted source code and design are my own work and that all external assets or references have been acknowledged appropriately.

Student Signature: W. Jalai Date: _____

Evaluator's Remarks

Strengths observed: _____		
Corrections / improvements required: _____		
Marks Awarded: _____ / 100	Evaluator Signature: _____	Date: _____

UNIT II – SCENARIO-BASED ASSESSMENT

Question 1 – Responsive University Event Registration Portal

Complete Answer up to Section H

The following answer follows the requirements in the question paper and is arranged in the same order from Section A to Section H.

A. Required Webpage Sections

The webpage contains a header with university name, portal title and navigation; a hero section with registration message and call-to-action; four event cards; a registration form; and a footer with contact details and useful links.

B. Mandatory CSS Requirements

1. Methods of Applying CSS: Inline CSS is used once for the small note at the bottom. Internal CSS is used for the announcement section. External CSS is used for the main website design. External CSS is the best choice for a multi-page website because one stylesheet can be reused across all pages.

2. Selectors: Element selector, class selector, ID selector, group selector, descendant selector, child selector, attribute selector, pseudo-class and pseudo-element are included. Hover and focus states are also provided.

3. Box Model: Every event card contains content, padding, border and margin. Example calculation: content width 250px + left/right padding 24px + left/right border 2px = **302px** total horizontal width under content-box.

Border-box: If width is set to 250px with box-sizing: border-box, the total width remains 250px because padding and border are included within the declared width.

4. Layout: CSS Grid is used for the event catalogue. Four columns are used on large screens, two on medium screens and one on small screens. The HTML order remains logical.

5. Positioning: Sticky navigation, relative event cards, absolute status badges, fixed help button and z-index are demonstrated. These elements are positioned so they do not cover important form fields or content.

6. Other CSS properties: The design uses background colours, gradients, border-radius, box-shadow, text-shadow, opacity, overflow, cursor, transition, transform, list-style and max-width.

7. Responsive design: Media queries change the navigation, event cards and registration form for smaller screens. At about 375px the cards appear in one column and horizontal scrolling is avoided.

C. Functional and Quality Expectations

Navigation links and buttons have clear labels and hover/focus feedback. The form uses labels and appropriate input types such as text and email. Alignment, spacing and typography are kept consistent. The layout is designed to remain readable at 125% browser zoom. CSS comments separate the major sections.

D. Suggested Open-Source Tools

Tool	Use
VSCodium / Code OSS	Create and edit HTML and CSS files
Mozilla Firefox	Primary browser testing and responsive inspection
Chromium	Cross-browser testing
Developer Tools	Inspect selectors, styles and box model
Live Server	Run the website locally
Git	Maintain project versions

E. Required Deliverables

1. index.html –complete semantic webpage structure.
2. style.css – main external CSS rules.
3. Desktop-view screenshot.
4. Mobile responsive screenshot.
5. CSS box-model screenshot for one event card.
6. One-page technical note explaining CSS methods, selectors, layout, positioning and responsive behaviour.
7. Total-width calculation for one event card.
8. Complete project folder containing all files and assets.

F. Required Testing Evidence

Test	Expected result	Status
Desktop view	Cards aligned in multiple columns	PASS
Mobile 375px	Cards become one column; no horizontal scroll	PASS
Navigation	Links open the correct sections	PASS
Buttons	Clear labels and hover feedback	PASS
Registration form	Fields and labels remain aligned	PASS
Firefox	Website displays correctly	PASS
Chromium	Website displays correctly	PASS

Layout Issue 1: Event cards became too narrow on smaller screens. **Correction:** Media queries change the grid from four columns to two and then one column.

Layout Issue 2: Navigation became crowded on mobile. **Correction:** The navigation changes to a column layout and links are allowed to wrap.

Developer Tools should be used to inspect computed styles and the box model before taking the final screenshot.

G. Suggested Time Allocation

Activity	Suggested time
HTML structure and semantic sections	10 minutes
Basic CSS, typography and box model	15 minutes
Flexbox or Grid layout	15 minutes
Positioning and additional visual properties	10 minutes
Responsive testing and corrections	10 minutes
Screenshots and short documentation	10 minutes
Recommended total	70 minutes

H. Student Submission Checklist

Submission item	Check
index.html is included and opens correctly.	■
style.css is linked correctly.	■
Desktop-view screenshot is included.	■
Mobile-view screenshot is included.	■
CSS box-model screenshot is included.	■
Technical note and width calculation are included.	■

Testing evidence from two browsers is included.	■
Complete project folder or repository export is submitted.	■

Source Code – index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>University Event Registration Portal</title>
<link rel="stylesheet" href="style.css">

<!-- Internal CSS -->
<style>
.announcement {
  background: linear-gradient(90deg, #17305f, #2864c7);
  color: white;
  text-align: center;
  padding: 12px;
  border-radius: 10px;
}
</style>
</head>

<body>

<header>
<nav class="navbar">
  <div class="logo">
    <h1>SIMATS University</h1>
    <p>Event Registration Portal</p>
  </div>

  <ul class="nav-links">
    <li><a href="#home">Home</a></li>
    <li><a href="#events">Events</a></li>
    <li><a href="#register">Register</a></li>
    <li><a href="#contact">Contact</a></li>
  </ul>
</nav>
</header>

<section id="home" class="hero">
  <div class="hero-content">
    <p class="tag">UNIVERSITY EVENTS 2026</p>
    <h2>Learn. Compete. Create.</h2>
    <p>Take part in technical, cultural, placement and sports events conducted by the university.</p>
    <a href="#register" class="main-button">Register Now</a>
  </div>
</section>

<div class="announcement">
  Registration is open from 8 August to 25 August 2026.
</div>

<section id="events" class="section">
  <div class="section-title">
    <p class="tag">EVENT CATALOGUE</p>
    <h2>Upcoming Events</h2>
  </div>

  <div class="event-container">

    <article class="event-card featured">
      <span class="status">Open</span>
      <div class="event-icon">■</div>
      <h3>AI and Web Hackathon</h3>
      <p class="category">Technical Event</p>
      <p><strong>Date:</strong> 18 August 2026</p>
      <p><strong>Time:</strong> 9:00 AM - 5:00 PM</p>
      <p><strong>Venue:</strong> Innovation Lab</p>
      <p>Develop an innovative website or AI solution and compete with other student teams.</p>
      <a href="#register" class="event-button">Register</a>
    </article>

    <article class="event-card">
      <span class="status">Open</span>
      <div class="event-icon">■</div>
      <h3>Campus Cultural Fest</h3>
      <p class="category">Cultural Event</p>
      <p><strong>Date:</strong> 20 August 2026</p>
      <p><strong>Time:</strong> 10:00 AM - 6:00 PM</p>
      <p><strong>Venue:</strong> Main Auditorium</p>
      <p>Students can participate in music, dance, drama and other cultural activities.</p>
      <a href="#register" class="event-button">Register</a>
    </article>

    <article class="event-card">
      <span class="status">Open</span>
      <div class="event-icon">■</div>
      <h3>Placement Skill Bootcamp</h3>
      <p class="category">Placement Event</p>
      <p><strong>Date:</strong> 22 August 2026</p>
      <p><strong>Time:</strong> 9:30 AM - 3:30 PM</p>
      <p><strong>Venue:</strong> Seminar Hall</p>
      <p>Improve aptitude, communication, interview and resume-building skills.</p>
      <a href="#register" class="event-button">Register</a>
    </article>

    <article class="event-card">
      <span class="status">Open</span>
      <div class="event-icon">■</div>
      <h3>Inter-Department Sports Meet</h3>
      <p class="category">Sports Event</p>
      <p><strong>Date:</strong> 25 August 2026</p>
      <p><strong>Time:</strong> 8:00 AM - 4:00 PM</p>
      <p><strong>Venue:</strong> University Ground</p>
      <p>Participate in football, volleyball, athletics and other sports competitions.</p>
      <a href="#register" class="event-button">Register</a>
    </article>

  </div>

</div>
```

```

        </article>

    </div>
</section>

<section id="register" class="section registration">
    <div class="section-title">
        <p class="tag">ONLINE REGISTRATION</p>
        <h2>Register for an Event</h2>
    </div>

    <form class="registration-form">

        <div class="form-group">
            <label for="studentName">Student Name</label>
            <input type="text" id="studentName" name="studentName"
                placeholder="Enter your name" required>
        </div>

        <div class="form-group">
            <label for="registerNumber">Register Number</label>
            <input type="text" id="registerNumber" name="registerNumber"
                placeholder="Enter register number" required>
        </div>

        <div class="form-group">
            <label for="department">Department</label>
            <select id="department" name="department" required>
                <option value="">Select Department</option>
                <option>Information Technology</option>
                <option>Computer Science</option>
                <option>Electronics</option>
                <option>Mechanical</option>
            </select>
        </div>

        <div class="form-group">
            <label for="email">Email Address</label>
            <input type="email" id="email" name="email"
                placeholder="Enter email address" required>
        </div>

        <div class="form-group full">
            <label for="event">Select Event</label>
            <select id="event" name="event" required>
                <option value="">Select an Event</option>
                <option>AI and Web Hackathon</option>
                <option>Campus Cultural Fest</option>
                <option>Placement Skill Bootcamp</option>
                <option>Inter-Department Sports Meet</option>
            </select>
        </div>

        <button type="submit" class="submit-button">
            Submit Registration
        </button>

    </form>
</section>

<a href="#contact" class="help-button" aria-label="Contact help"></a>

<footer id="contact">
    <div class="footer-column">
        <h2>SIMATS University</h2>
        <p>Event Registration Portal</p>
        <p>Email: events@simats.example</p>
        <p>Phone: +91 90000 00000</p>
    </div>

    <div class="footer-column">
        <h3>Useful Links</h3>
        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#events">Events</a></li>
            <li><a href="#register">Register</a></li>
        </ul>
    </div>

    <p class="copyright">
        © 2026 SIMATS University. All Rights Reserved.
    </p>
</footer>

<!-- Inline CSS -->
<p style="text-align:center; font-size:12px; opacity:0.7;">
    Designed for Web Technology Unit II.
</p>

</body>
</html>

```


Source Code – style.css

```
/* =====  
UNIVERSITY EVENT REGISTRATION PORTAL  
External CSS  
===== */  
  
* {  
  box-sizing: border-box;  
  margin: 0;  
  padding: 0;  
}  
  
html {  
  scroll-behavior: smooth;  
}  
  
body {  
  font-family: Arial, Helvetica, sans-serif;  
  background: #f4f7fb;  
  color: #172033;  
  line-height: 1.6;  
  overflow-x: hidden;  
}  
  
/* Navigation */  
.navbar {  
  position: sticky;  
  top: 0;  
  z-index: 1000;  
  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  gap: 20px;  
  
  padding: 16px 6%;  
  background: white;  
  
  box-shadow: 0 3px 15px rgba(0, 0, 0, 0.10);  
}  
  
.logo h1 {  
  color: #1d4ed8;  
  font-size: 24px;  
}  
  
.logo p {  
  color: #667085;  
  font-size: 13px;  
}  
  
.nav-links {  
  display: flex;  
  gap: 25px;  
  list-style: none;  
}  
  
.nav-links a {  
  color: #172033;  
  text-decoration: none;  
  font-weight: bold;  
  transition: 0.3s;  
}  
  
.nav-links a:hover,  
.nav-links a:focus {  
  color: #2563eb;  
}  
  
/* Hero */  
.hero {  
  min-height: 430px;  
  display: flex;  
  align-items: center;  
  padding: 70px 6%;  
  background: linear-gradient(120deg, #172554, #2563eb);  
  color: white;  
}  
  
.hero-content {  
  max-width: 750px;  
}  
  
.tag {  
  letter-spacing: 2px;  
  font-size: 12px;  
  font-weight: bold;  
  opacity: 0.85;  
}  
  
.hero h2 {  
  font-size: clamp(40px, 7vw, 70px);  
  line-height: 1.1;  
  margin: 15px 0;  
  text-shadow: 2px 3px 10px rgba(0, 0, 0, 0.25);  
}  
  
.hero-content > p {  
  max-width: 650px;  
}  
  
/* Buttons */  
.main-button,  
.event-button,  
.submit-button {  
  display: inline-block;
```

```

margin-top: 20px;
padding: 11px 20px;
border: none;
border-radius: 9px;
background: #f59e0b;
color: #111827;
text-decoration: none;
font-weight: bold;
cursor: pointer;
transition: transform 0.2s ease, box-shadow 0.2s ease;
}

.main-button:hover,
.event-button:hover,
.submit-button:hover {
  transform: translateY(-3px);
  box-shadow: 0 8px 18px rgba(0, 0, 0, 0.20);
}

/* Announcement */
.announcement {
  max-width: 1100px;
  margin: 20px auto;
}

/* Sections */
.section {
  max-width: 1180px;
  margin: auto;
  padding: 60px 6%;
}

.section-title {
  margin-bottom: 30px;
}

.section-title h2 {
  font-size: 32px;
}

/* Event Grid */
.event-container {
  display: grid;
  grid-template-columns: repeat(4, minmax(0, 1fr));
  gap: 22px;
}

/* Event Card */
.event-card {
  position: relative;
  padding: 24px;
  margin: 4px;
  border: 2px solid #dbe3f0;
  border-radius: 16px;
  background: white;
  box-shadow: 0 8px 25px rgba(30, 41, 59, 0.08);
  overflow: hidden;
  transition: transform 0.2s ease, box-shadow 0.2s ease;
}

.event-card:hover {
  transform: translateY(-6px);
  box-shadow: 0 15px 30px rgba(30, 41, 59, 0.14);
}

/* Pseudo-element */
.event-card::before {
  content: "";
  position: absolute;
  left: 0;
  top: 0;
  width: 5px;
  height: 100%;
  background: #2563eb;
}

/* Child selector */
.event-card > h3 {
  margin-top: 10px;
  color: #172554;
}

.event-card.featured {
  border-color: #2563eb;
}

.status {
  position: absolute;
  top: 15px;
  right: 15px;

  padding: 4px 10px;
  border-radius: 20px;

  background: #dcfce7;
  color: #166534;

  font-size: 12px;
  font-weight: bold;
}

.event-icon {
  font-size: 35px;
}

.category {
  color: #2563eb;
  font-weight: bold;
}

```

```

    margin-bottom: 10px;
}

.event-button {
    background: #2563eb;
    color: white;
    padding: 8px 16px;
    margin-top: 12px;
}

/* Registration Form */
.registration {
    max-width: 900px;
}

.registration-form {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 18px;

    padding: 28px;
    border-radius: 16px;
    background: white;

    box-shadow: 0 8px 25px rgba(30, 41, 59, 0.08);
}

.form-group {
    display: flex;
    flex-direction: column;
    gap: 6px;
}

.form-group.full {
    grid-column: 1 / -1;
}

label {
    font-weight: bold;
}

input,
select {
    width: 100%;
    padding: 11px 12px;

    border: 1px solid #b9c5d6;
    border-radius: 8px;

    font: inherit;
    background: white;
}

/* Focus pseudo-class */
input:focus,
select:focus {
    outline: 3px solid rgba(37, 99, 235, 0.20);
    border-color: #2563eb;
}

/* Attribute selector */
input[type="email"] {
    background: #f8fbff;
}

.submit-button {
    grid-column: 1 / -1;
    justify-self: start;

    background: #172554;
    color: white;
}

/* Fixed positioning */
.help-button {
    position: fixed;
    right: 22px;
    bottom: 22px;
    z-index: 900;

    width: 48px;
    height: 48px;

    display: grid;
    place-items: center;

    border-radius: 50%;
    background: #f59e0b;
    color: #111827;

    text-decoration: none;
    font-size: 22px;
    font-weight: bold;

    box-shadow: 0 8px 18px rgba(0, 0, 0, 0.20);
}

/* Footer */
footer {
    padding: 40px 6%;
    background: #111827;
    color: #e5e7eb;

    display: flex;
    flex-wrap: wrap;
    justify-content: space-between;
    gap: 25px;
}

```

```

footer a {
  color: #e5e7eb;
  text-decoration: none;
}

footer a:hover {
  color: #60a5fa;
}

footer ul {
  list-style: square;
  padding-left: 20px;
}

.copyright {
  flex-basis: 100%;
  opacity: 0.75;
}

/* Responsive Design */
@media(max-width: 900px) {
  .event-container {
    grid-template-columns: repeat(2, minmax(0, 1fr));
  }
}

@media (max-width: 600px) {

  .navbar {
    flex-direction: column;
    align-items: flex-start;
  }

  .nav-links {
    width: 100%;
    flex-wrap: wrap;
    gap: 10px 16px;
  }

  .hero {
    min-height: 360px;
    padding: 50px 6%;
  }

  .event-container {
    grid-template-columns: 1fr;
  }

  .registration-form {
    grid-template-columns: 1fr;
  }

  .form-group.full,
  .submit-button {
    grid-column: auto;
  }

  .section {
    padding: 45px 5%;
  }

  .help-button {
    right: 14px;
    bottom: 14px;
  }
}

```